NIOS Class 10 (212) Science & Technology — Chapter-by-Chapter Fast Revision

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

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> **Only the 19 Public Examination Chapters are summarized here.**
> The 13 TMA chapters have been excluded so your study time is 100% focused on what will be tested.

MODULE 1: MEASUREMENT IN SCIENCE

Lesson 1: Measurement in Science & Technology

- **Fundamental Quantities & SI Units**: Length (m), Mass (kg), Time (s), Temperature (K), Electric Current (A), Luminous Intensity (cd), Amount of Substance (mol).
- **Derived Quantities**: Area (m^2), Volume (m^3), Density ($\mathrm{kg/m}^3$), Speed ($\mathrm{m/s}$), Force (N).
- **Key Instruments**: Vernier Caliper (least count $0.01\ \mathrm{cm} = 0.1\ \mathrm{mm}$), Screw Gauge / Micrometer (least count $0.01\ \mathrm{mm}$).

MODULE 2: MATTER IN OUR SURROUNDINGS (CHEMISTRY)

Lesson 5: Atomic Structure

- **Subatomic Particles**: Electron (-1, Thomson), Proton ($+1$, Rutherford), Neutron (0, Chadwick).
- **Bohr's Model**: Electrons revolve in discrete circular orbits (energy shells K, L, M, N with maximum $2n^2$ electrons: $2, 8, 18, 32$).
- **Atomic Number (Z)** = Number of protons.

- **Mass Number (A)** = Protons + Neutrons.
- **Isotopes vs Isobars**:
- *Isotopes*: Same atomic number Z, different mass number A (e.g. ${}^{12}_{6}\mathrm{C}$ and ${}^{14}_{6}\mathrm{C}$; ${}^{1}_{1}\mathrm{H}$, ${}^{2}_{1}\mathrm{H}$, ${}^{3}_{1}\mathrm{H}$).
- *Isobars*: Same mass number A, different atomic number Z (e.g. ${}^{40}_{18}\mathrm{Ar}$ and ${}^{40}_{20}\mathrm{Ca}$).

Lesson 6: Periodic Classification of Elements

- **Modern Periodic Law**: Physical and chemical properties of elements are periodic functions of their **atomic numbers**.
- **Modern Periodic Table Layout**: 18 vertical columns (**Groups**) and 7 horizontal rows (**Periods**).
- **Periodic Trends**:
- *Valency*: Across period: increases from 1 to 4, then decreases to 0. Down a group: remains constant.
- *Atomic Size (Radius)*: **Decreases** left to right across a period (due to increasing nuclear charge pulling electrons closer); **Increases** down a group (due to addition of new electron shells).
- *Metallic Character*: Decreases across a period; increases down a group.

Lesson 7: Chemical Bonding

- **Octet Rule**: Atoms gain, lose, or share valence electrons to attain a stable noble gas configuration (8 electrons in outermost shell).
- **Ionic (Electrovalent) Bond**: Formed by complete transfer of electrons from a metal to a non-metal (e.g., $\mathrm{NaCl}, \mathrm{MgCl_2}$). High melting points, soluble in water, conduct electricity in molten/aqueous state.
- **Covalent Bond**: Formed by mutual sharing of electron pairs between non-metals (e.g., $\mathrm{H_2}, \mathrm{O_2}, \mathrm{CH_4}, \mathrm{H_2O}$). Lower melting points, non-conductors of electricity.

Lesson 8: Acids, Bases and Salts

- **Acids**: Taste sour, produce $\mathrm{H^+}$ ions in water, turn blue litmus red.
- **Bases**: Taste bitter, soapy to touch, produce $\mathrm{OH^-}$ ions in water, turn red litmus blue.
- **Salts**: Formed by neutralization: $\mathrm{Acid + Base \rightarrow Salt + Water}$.
- **Master Equations**:
- Bleaching Powder: $\mathrm{Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O}$
- Baking Soda: $\mathrm{NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3}$
- POP: $\mathrm{CaSO_4 \cdot 2H_2O \xrightarrow{373\ K} CaSO_4 \cdot \frac{1}{2}H_2O + 1.5H_2O}$

MODULE 3: MOVING THINGS (PHYSICS)

Lesson 10: Force and Motion

- **Newton's First Law**: A body continues in state of rest or uniform motion unless acted upon by an external unbalanced force (Law of Inertia).
- **Newton's Second Law**: Rate of change of momentum is directly proportional to applied force: $F = ma$.
- **Newton's Third Law**: To every action, there is an equal and opposite reaction acting on two different bodies.
- **Law of Conservation of Momentum**: In an isolated system, total momentum before collision equals total momentum after collision: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.

Lesson 11: Gravitation

- **Universal Law of Gravitation**: $F = G\frac{Mm}{d^2}$ ($G = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$).
- **Acceleration due to Gravity (g)**: $g = \frac{GM}{R^2} \approx 9.8\ \mathrm{m/s^2}$ on Earth; $g_{\mathrm{moon}} = \frac{1}{6}g_{\mathrm{earth}}$.
- **Difference between Mass and Weight**:
- *Mass*: Quantity of matter; constant everywhere; scalar; measured in kg.
- *Weight*: Gravitational force $W = mg$; varies with location; vector; measured in N.
- **Archimedes' Principle**: An immersed body experiences an upward buoyant force equal to the weight of fluid displaced.

MODULE 4: ENERGY (PHYSICS)

Lesson 13: Work and Energy

- **Work**: $W = F \cdot s$ (Joules). No work is done if force is perpendicular to displacement ($\cos 90^\circ = 0$).
- **Kinetic Energy**: $KE = \frac{1}{2}mv^2$.
- **Potential Energy**: $PE = mgh$.
- **Law of Conservation of Energy**: Energy can neither be created nor destroyed; only converted from one form to another. Total mechanical energy $E = KE + PE$ remains constant.
- **Power**: $P = \frac{W}{t}$ (Watts). $1\ \mathrm{kW} = 1000\ \mathrm{W}$.

Lesson 15: Light Energy

- **Reflection**: Angle of incidence $=$ Angle of reflection ($\angle i = \angle r$).
- **Mirror Formula**: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$, Magnification $m = -\frac{v}{u}$.
- **Refraction & Snell's Law**: $\mu = \frac{\sin i}{\sin r}$. Light bends towards normal entering a denser medium.
- **Lens Formula**: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$, Magnification $m = +\frac{v}{u}$.
- **Power of Lens**: $P = \frac{1}{f\,(\mathrm{m})}$ (Dioptres). Convex is $+$, Concave is $-$.
- **Eye Defects**: Myopia (corrected by concave lens) vs Hypermetropia (corrected by convex lens).

Lesson 16: Electrical Energy

- **Ohm's Law**: $V = IR$.
- **Resistance Factors**: $R = \rho\frac{l}{A}$.
- **Resistors**: Series $R_s = R_1 + R_2$; Parallel $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$.
- **Joule's Heating**: $H = I^2Rt$.
- **Commercial Energy**: $1\ \mathrm{kWh} = 3.6 \times 10^6\ \mathrm{J} = 1\ \mathrm{Unit}$.

Lesson 17: Magnetic Effect of Electric Current

- **Right-Hand Thumb Rule**: Thumb indicates current direction; curled fingers indicate circular magnetic field lines.
- **Solenoid**: Coil of insulated wire; magnetic field inside a long current-carrying solenoid is uniform and strong (behaves like a bar magnet).
- **Fleming's Left-Hand Rule**: Used in electric motors (Thumb = Motion, Index = Field, Middle = Current).
- **Safety Devices**: Earth wire (diverts leakage current safely to ground), Fuse (melts and breaks circuit during overload/short-circuit).

MODULE 5: THE LIVING WORLD (BIOLOGY)

Lesson 22: Life Processes – I (Nutrition, Respiration, Circulation, Excretion)

- **Photosynthesis**: Conversion of solar energy to chemical energy in green plants using chlorophyll.
- **Human Digestion**: Salivary amylase in mouth, Pepsin & HCl in stomach, Bile & Pancreatic juice (trypsin, lipase) in small intestine, absorption through villi.

- **Respiration**: Breakdown of glucose: Aerobic (mitochondria, 38 ATP) vs Anaerobic (yeast: ethanol; muscle: lactic acid, 2 ATP).
- **Circulation**: 4-chambered human heart prevents mixing of oxygenated and deoxygenated blood; Double circulation.
- **Excretion**: Nephron filters blood in kidneys; 3 steps: Ultrafiltration $\rightarrow$ Selective Reabsorption $\rightarrow$ Tubular Secretion.

Lesson 23: Life Processes – II (Control & Coordination)

- **Neuron Structure**: Dendrite $\rightarrow$ Cyton (cell body) $\rightarrow$ Axon $\rightarrow$ Nerve ending $\rightarrow$ Synapse (chemical transmission via neurotransmitter).
- **Reflex Arc**: Receptor $\rightarrow$ Sensory neuron $\rightarrow$ Spinal cord $\rightarrow$ Motor neuron $\rightarrow$ Effector muscle.
- **Plant Hormones (Phytohormones)**:
- *Auxin*: Shoot elongation and phototropism.
- *Gibberellin*: Stem growth.
- *Cytokinin*: Promotes cell division.
- *Abscisic Acid (ABA)*: Growth inhibitor, wilting of leaves.
- **Endocrine Glands**: Pancreas (Insulin), Thyroid (Thyroxine requires iodine), Adrenal (Adrenaline - fight/flight hormone).

Lesson 24: Life Processes – III (Reproduction)

- **Asexual Reproduction**: Binary fission (Amoeba), Budding (Yeast, Hydra), Spore formation (Rhizopus), Fragmentation (Spirogyra), Vegetative propagation (Potato, Rose).
- **Sexual Reproduction in Plants**: Flower parts (Stamen = male: anther + filament; Carpel = female: stigma + style + ovary). Pollination (self/cross) followed by fertilization.
- **Human Reproduction**: Testes produce sperm and testosterone; Ovaries produce ova, estrogen, and progesterone. Fertilization occurs in Fallopian tube.

Lesson 25: Heredity

- **Mendel's Laws**: Law of Segregation and Law of Independent Assortment.
- **Monohybrid Cross**: Phenotypic ratio $3:1$, Genotypic ratio $1:2:1$.
- **Sex Determination in Humans**: $44+XX$ (female) and $44+XY$ (male). Father determines child's sex.

MODULE 6: NATURAL RESOURCES (CHEMISTRY)

Lesson 27: Metals and Non-metals

- **Reactivity Series**: $\mathrm{K} > \mathrm{Na} > \mathrm{Ca} > \cdots > \mathrm{Cu} > \mathrm{Ag} > \mathrm{Au}$.
- **Extraction of Metals**:
- Sulphide ores $\rightarrow$ Roasting (excess air).
- Carbonate ores $\rightarrow$ Calcination (limited air).
- **Corrosion**: Oxidation of metals exposed to moisture and air. Prevention: Galvanization (coating iron with zinc), painting, greasing, making alloys (e.g. Stainless steel: $\mathrm{Fe} + \mathrm{Cr} + \mathrm{Ni}$).

Lesson 28: Carbon and Its Compounds

- **Catenation & Tetravalency**: Give rise to huge variety of carbon compounds.
- **Homologous Series**: Family of organic compounds having same functional group and general formula; successive members differ by $-\mathrm{CH_2}-$ unit and $14\ \mathrm{u}$ molecular mass.
- **Soaps and Detergents**: Soaps form micelles to remove grease; ineffective in hard water due to scum formation ($\mathrm{Ca}^{2+}/\mathrm{Mg}^{2+}$ precipitates). Synthetic detergents work effectively even in hard water.

MODULE 7: HUMAN AND ENVIRONMENT (ECOLOGY)

Lesson 29: Natural Environment

- **Ecosystem**: Functional unit comprising biotic components (plants, animals, decomposers) and abiotic components (air, water, soil, sunlight).
- **Food Chain & Food Web**: Linear flow of energy vs interconnected network of food chains.
- **10% Law of Energy**: Only 10% of energy transfers to the next trophic level; limits food chains to 3–4 trophic levels.

Lesson 30: Human Impact on Environment

- **Ozone Layer Depletion**: $\mathrm{O_3}$ layer in stratosphere shields Earth from UV rays. Depleted by CFCs from cooling appliances.
- **Biomagnification**: Progressive accumulation of toxic, non-biodegradable chemicals (like DDT) along food chains.

- **Waste Management**: Segregation into Biodegradable (organic, decomposes naturally) and Non-biodegradable (plastics, glass; requires 3Rs: Reduce, Reuse, Recycle).